

module-11-multiplication-near-base - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 11 — Multiplication Near a Power of 10 (Darśana)

Sanskrit: *Nikhilam* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Fast multiplication when numbers are near 10, 100, or 1000.

Darśana note. This workshop condenses Module 11's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Project $97 \times 96 = ?$ “Hard to do in your head — usually. But there's a trick: cross-subtract and multiply deficits. $97 \times 96 = 9312$. Today's method: *Nikhilam for multiplication*.”

Core idea · 15 min

Nikhilam multiplication for numbers near a power of 10:

For $(B - a)(B - b)$, where B is a power of 10: 1. Compute deficits: a, b . 2. Cross-subtract: $(B - a) - b$ OR $(B - b) - a$ (same answer). 3. Multiply deficits: $a \times b$. 4. Result: [cross-subtract] [multiply], with appropriate padding.

Example: 97×96 . - Deficits: 3, 4. - Cross-subtract: $97 - 4 = 93$. Or $96 - 3 = 93$. - Multiply deficits: $3 \times 4 = 12$. - Answer: 9312.

Numbers above base: 103×104 — excesses 3, 4; cross-add $\rightarrow 107$; multiply $\rightarrow 12$; answer 10712.

Demonstration · 10 min

Live speed: 5 problems near 100 (e.g., 96×93 , 88×89). Each student solves mentally. Time it.

Source moment · 5 min

Paraphrase Tirthaji 1965, Sūtra 2 (same as Module 9, applied to multiplication): “*Nikhilam — all from 9, last from 10 — gives the deficits, which then drive the multiplication.*”

Hands-on activity · 15 min

Mental-math sprint. 15 problems of near-100 multiplication. Each student records time + accuracy. Pair-share strategies.

Reflection + take-home · 10 min

“*For what % of multiplication problems would Nikhilam help? When does it NOT help?*”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Finger-trick prelude
- **Day 02** — 97×96 — two below base
- **Day 03** — 103×104 — two above base
- **Day 04** — Mixed cases
- **Day 05** — 3-digit near 1000
- **Day 06** — Algebra proof
- **Day 07** — When to use
- **Day 08** — Speed sprints
- **Day 09** — Mixed practice — combining $\times 11$ and Nikhilam
- **Day 10** — Final assessment

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.